

PARTH AGARWAL

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Education

The Pennsylvania State University

B.S. Computer Science, Mathematics

Expected Graduation: May 2027

Cumulative GPA: 3.57/4.00

Coursework: Linear Algebra, Discrete Math, Data Structures, Object Oriented Programming, ML (Machine learning) & AI (Artificial Intelligence)

Technical Skills

Programming/Software: C, C++, Python, Java, MATLAB, SQL, Linux, Git, GitHub, ROS/ROS 2, SolidWorks, PyTorch, TensorFlow, scikit-learn, NumPy, pandas, Jupyter, LaTeX, Bash, Docker

Research Experience

Undergraduate Research— PI: Prof. Haohan Wang, UIUC (Verifiable Agentic AI for Computational Biology) May 2026— Present

- **Conference Poster:** *A General Biomedical Agentic Framework* — accepted and presented at the **Midwest Machine Learning Symposium (MMLS 2026)**, Purdue University, June 2026.
- Researching a compiler-inspired biomedical agent architecture that converts open-ended scientific intent into typed plans, registry-bound primitives, deterministic execution, and verifier-authorized results.
- Designing semantic biological-role resolution and typed execution contracts for matrices, cohorts, conditions, covariates, annotations, and parameters across H5AD, MTX, VCF, and BED inputs.
- Building a reusable scientific capability layer across single-cell, transcriptomics, and genomics, exposing pseudobulk DE, GENIE3/GRNBoost2, WGCNA, enrichment, variant/peak annotation, and motif disruption.
- Developing independent scientific adjudication and recovery using verifier-side recomputation, provenance/hash invariants, adversarial tests, repeat-parity checks, and fail-closed authorization.

Undergraduate Research— PI: Prof. Fenglong Ma, PSU (Multimodal EHR Modeling + Clinical NLP) Sep 2025 — Aug 2026

- Building a modular pipeline for EHR time-series and clinical text with standardized schemas, dataloaders, and explicit missing-modality handling (dropout/imputation).
- Implementing a masked-attention fusion transformer with per-modality encoders and a clean config/trainer API; evaluating against strong single-modality and late/early-fusion baselines with fixed protocols (AUROC/AUPRC, calibration).
- Delivering an analysis suite: per-modality attribution and token/feature-level explanations (attention rollout, SHAP), robustness checks under missingness/shift, and a concise technical report/poster with ablations.

Undergraduate Research - PI: Prof. Suman Saha, PSU Nov 2024 — August 2025

Explainable Gait Abnormality Detection Using Dual-Dataset CNN-LSTM Model— [arXiv:2509.16472](https://arxiv.org/abs/2509.16472)

- Designed and trained a hybrid CNN-LSTM for cross-view gait recognition on OU-MVLP and GAVD.
- Built KNN/DT/RF and deep models from scratch; full PyTorch training pipeline.
- Preprocessed sequences; handled imbalance with SMOTE + loss weighting.
- Validated on unseen identities to demonstrate robustness for deployment.

Undergraduate Research Volunteer — Systems and Network Security Lab(SyNSec) Dec 2023 — Aug 2025

Serving as a research volunteer in the Department of Computer Science at Penn State under Dr. Syed Rafiul Hussain

Recognized Contributing to Research to Award Number: 2215017

Project Title: Collaborative Research: CNS Core: Large: Systems and Verifiable Metrics for Sustainable Data Centers
Simulated and executed fuzzing tools using C++ such as AFL, AFL ++, AFLNET, LibAFL to check for vulnerabilities and deviation in 5G networks

- Crafted a LibAFL code in C++ and Python via Virtual Machine as a building block for future development on projects requiring NLP (Natural Language Processing)
- Partnered with cross-functional teams to integrate and test security solutions, significantly enhancing the security and privacy of cellular and IoT systems.

Technical Experience

Guidance, Navigation, and Controls — Penn State Advanced Vehicle Team

Dec 2023 — Aug 2025

*SAE AutoDrive Challenge II — Sponsored by **General Motors & SAE International***

- Contributed to the development of an autonomous vehicle for the GM–SAE AutoDrive Challenge II, targeting SAE Level 4 autonomy in complex urban driving environments.
- Achieved 3rd place overall out of 10 teams in Auto Drive Challenge II Year 3 Michigan June 2024
- Achieved 2nd place overall on Intersection Challenge in Auto Drive Challenge II Year 3 Michigan June 2024
- Achieved 2nd place overall on Construction Challenge in Auto Drive Challenge II Year 3 Michigan June 2024

- Engineered an NVIDIA Jetson-based HMI using MATLAB, Python, Java/JavaScript, and ROS to interface with autonomous-vehicle subsystems.
- Advanced the development of high-definition (HD) maps, fostering the refinement of algorithms geared towards optimizing car navigation efficiency while concurrently advancing comprehension of the vehicle's precise spatial orientation and operational modalities

Solutions Engineer – Penn State Nittany AI Advance

May 2025 – Dec 2025

- Built an end-to-end student intake and advising case workflow for Penn State Undergraduate Education using Azure and Power Automate, including routing logic, status tracking, and automated notifications for advisors.
- Developed Python scripts for data validation, cleaning, and structured record generation to improve consistency and reduce manual back and forth in the advising process.
- Delivered a maintainable solution through iterative stakeholder demos, testing, and documentation, improving advisor visibility and streamlining request handling.

Selected Publications & Presentations

1. **Parth Agarwal**, Sangaa Chatterjee, Md Faisal Kabir, and Suman Saha. *Explainable Gait Abnormality Detection Using Dual-Dataset CNN-LSTM Model*. In Proc. of the IEEE International Conference on Machine Learning and Applications (ICMLA), 2025. [arXiv:2509.16472](https://arxiv.org/abs/2509.16472)
2. **Parth Agarwal** and Haohan Wang. *A General Biomedical Agentic Framework*. Poster presented at the Midwest Machine Learning Symposium (MMLS), Purdue University, 2026.

Research Fellowships

College of Engineering Research Funding – PSU

Sep 2026 – May 2027

- Selected for a highly competitive, donor-funded undergraduate research award from the Penn State College of Engineering, providing \$4,000 in research support across the Fall 2026 and Spring 2027 semesters.
- Awarded funding in recognition of the strength and potential impact of my ongoing undergraduate research, supporting continued research contributions throughout the 2026–2027 academic year.

Summer Undergraduate Research Experience (SURE) - PSU (Eberly College of Science)

May 2026 – July 2026

- Selected for a funded, 10-week full-time research fellowship, pursuing faculty-mentored multimodal clinical ML research under *Prof. Fenglong Ma*.
- Led a focused clinical ML research investigation, integrating literature synthesis, hypothesis formulation, controlled experimentation, ablation/error analysis, and development of subsequent research directions.

Professional Service

Program Committee Member, IEEE International Conference on Tools with Artificial Intelligence (ICTAI 2026)

Teaching & Academic Service

Teaching Assistant – DS 420 (Network Analytics)

Aug 2026 – Present

- Supported instruction and student problem-solving in graph and network analytics, helping students implement and interpret community detection, information propagation, node/link classification, and network embedding methods across real-world network applications.

Teaching Assistant – IST 242 (Intermediate & Object-Oriented Application Development)

Aug 2026 - Present

- Supported instruction and student debugging in intermediate application development, including algorithms, data structures, object-oriented programming, reusable code libraries, and APIs.

Learning Assistant – DS 200 (Introduction to Data Science)

Jan 2026 – May 2026

Supported instructor-led lectures and hands-on learning in an introductory data science course, reinforcing core concepts in data analysis, programming, visualization, and statistical reasoning.

- Facilitated weekly learning sessions on data wrangling, visualization, exploratory analysis, and introductory statistical/modeling concepts.
- Provided individualized support with Python/R workflows, debugging, data interpretation, and course assignments, strengthening students' computational and analytical reasoning.
- Delivered timely feedback during class and office hours and facilitated collaborative problem-solving to reinforce data science concepts and programming practices.

Math Peer Tutor – Penn State Learning Resource Center

Aug 2025 – Present

- Provided 1:1 and small-group tutoring for Calculus I/II, Sequences & Series, and intro proofs, reinforcing core problem-solving techniques.

Grader – CMPSC 465 (Data Structures and Algorithms)

Aug 2025 – Dec 2025

- Graded programming assignments and written problem sets on asymptotics, graph algorithms, dynamic programming, and NP-completeness using a detailed rubric.
- Delivered timely, actionable feedback via Canvas and clarified common errors to improve student performance.
- Maintained accurate gradebooks and flagged academic-integrity concerns per department.

Learning Assistant – CMPSC 132 (Programming and Computation II: Data Structures)

Dec 2023 – Aug 2024

- Facilitated dynamic sessions for 30+ students, breaking down complex Data Structures concepts.
- Elevated student performance with personalized mentorship, boosting coding skills.
- Enhanced assessments by delivering targeted feedback, driving improvement and success.

Learning Assistant – Math 141 (Calculus with Analytic Geometry II)

Dec 2023 – Aug 2024

Support instructor-led lectures and provide academic assistance to students in a large lecture-based calculus course. Facilitate peer learning, reinforce core mathematical concepts, and encourage collaborative problem-solving.

- Assisted students with understanding calculus topics such as integration techniques, sequences and series, and applications of the integral
- Facilitated small-group discussions during class and held weekly office hours to support student learning
- Collaborated with faculty and fellow LAs to develop effective teaching strategies and review sessions
- Encouraged student participation and fostered an inclusive and supportive learning environment